06 - zz - Dynamic Routing & BGP - EXCERCISE (my solution)

Lesson 4 Exercises: Dynamic Routing with BGP

Complete these exercises to understand how BGP replaces static routes with dynamic, self-healing routing.

Exercise 1: Deploy and Configure eBGP

Objective: Deploy the topology, observe that routers have no routes beyond their directly connected subnets, then configure eBGP to restore full connectivity.

Steps

1. Deploy the topology:
   ```
   cd lessons/clab/04-dynamic-routing-bgp
   containerlab deploy -t topology/lab.clab.yml
   ```
2. Install gNMIc if needed:
   ```
   brew install gnmic
   ```
3. Verify cross-subnet pings FAIL -- routers only know their directly connected subnets:
   ```
   docker exec clab-dynamic-routing-bgp-host1 ping -c 2 -W 3 10.1.4.2
   docker exec clab-dynamic-routing-bgp-host1 ping -c 2 -W 3 10.1.5.2
   ```
4. Check the routing table on srl1 -- only `local` routes for directly connected interfaces:
   ```
   docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli -c "show network-instance default route-table ipv4-unicast summary"
   ```
5. Examine the pre-wired but unconfigured srl2-srl3 link:
   ```
   docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli -c "show interface ethernet-1/3"
   ```

6. Apply BGP config using gNMIc:

```
cd gnmic
gnmic -a clab-dynamic-routing-bgp-srl1:57400 -u admin -p NokiaSrl1! \
  --skip-verify -e json_ietf set --request-file configs/srl1-bgp.json
gnmic -a clab-dynamic-routing-bgp-srl2:57400 -u admin -p NokiaSrl1! \
  --skip-verify -e json_ietf set --request-file configs/srl2-bgp.json
gnmic -a clab-dynamic-routing-bgp-srl3:57400 -u admin -p NokiaSrl1! \
  --skip-verify -e json_ietf set --request-file configs/srl3-bgp.json
```

7. Verify BGP sessions are established:

```
docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli -c "show network-
instance default protocols bgp neighbor"
```

8. Verify cross-subnet pings now work:

```
docker exec clab-dynamic-routing-bgp-host1 ping -c 3 10.1.4.2
docker exec clab-dynamic-routing-bgp-host1 ping -c 3 10.1.5.2
docker exec clab-dynamic-routing-bgp-host2 ping -c 3 10.1.5.2
```

9. Check the routing table -- remote subnets now show as `bgp`:

```
docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli -c "show network-
instance default route-table ipv4-unicast summary"
```

Deliverables

- Routing table before BGP (only local routes) and after (local + bgp routes)
- All cross-subnet pings succeeding

✅ done while following the video

Exercise 2: Enable the Direct Link and Observe Path Selection

Objective: Enable the pre-wired srl2-srl3 link and observe BGP selecting the shorter AS path.

Steps

1. Traceroute BEFORE the new link -- host2 to host3 goes through the hub (3 router hops):

```
docker exec clab-dynamic-routing-bgp-host2 traceroute -n -w 2 10.1.5.2
```

Expected: 10.1.4.1 (srl2) -> 10.1.2.1 (srl1) -> 10.1.3.2... -> 10.1.5.2

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-kubecraft/lessons/clab/04-dynamic-routing-bgp]
└──>  docker exec clab-dynamic-routing-bgp-host2 traceroute -n -w 2 10.1.5.2
traceroute to 10.1.5.2 (10.1.5.2), 30 hops max, 60 byte packets
1  10.1.4.1  1.022 ms  0.915 ms  0.900 ms
2  10.1.2.1  0.531 ms  1.317 ms  1.319 ms
3  10.1.3.2  1.091 ms  1.087 ms  1.426 ms
4  10.1.5.2  0.913 ms  0.932 ms  0.929 ms
```

2. Configure the srl2-srl3 interfaces:

```
gnmic -a clab-dynamic-routing-bgp-srl2:57400 -u admin -p NokiaSrl1! \
  --skip-verify -e json_ietf set --request-file configs/srl2-new-link.json
gnmic -a clab-dynamic-routing-bgp-srl3:57400 -u admin -p NokiaSrl1! \
  --skip-verify -e json_ietf set --request-file configs/srl3-new-link.json
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> gnmic -a clab-dynamic-routing-bgp-srl2:57400 -u admin -p NokiaSrl1!
--skip-verify -e json_ietf set --request-file conf
igs/srl2-new-link.json
{
  "source": "clab-dynamic-routing-bgp-srl2:57400",
  "timestamp": 1779672924477344259,
  "time": "2026-05-25T03:35:24.477344259+02:00",
  "results": [
    {
      "operation": "UPDATE",
      "path": "interface[name=ethernet-1/3]"
    },
    {
      "operation": "UPDATE",
      "path": "network-instance[name=default]/interface[name=ethernet-1/3.0]"
    }
  ]
}

[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> gnmic -a clab-dynamic-routing-bgp-srl3:57400 -u admin -p NokiaSrl1! \
    --skip-verify -e json_ietf set --request-file configs/srl3-new-link.json
{
  "source": "clab-dynamic-routing-bgp-srl3:57400",
  "timestamp": 1779672936182298889,
  "time": "2026-05-25T03:35:36.182298889+02:00",
  "results": [
    {
      "operation": "UPDATE",
      "path": "interface[name=ethernet-1/3]"
    },
    {
      "operation": "UPDATE",
      "path": "network-instance[name=default]/interface[name=ethernet-1/3.0]"
    }
  ]
}
```

3. Add BGP neighbors for the new link:

```
gnmic -a clab-dynamic-routing-bgp-srl2:57400 -u admin -p NokiaSrl1! \
  --skip-verify -e json_ietf set --request-file configs/srl2-bgp-
srl3.json
gnmic -a clab-dynamic-routing-bgp-srl3:57400 -u admin -p NokiaSrl1! \
  --skip-verify -e json_ietf set --request-file configs/srl3-bgp-
srl2.json
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> gnmic -a clab-dynamic-routing-bgp-srl2:57400 -u admin -p NokiaSrl1! \
    --skip-verify -e json_ietf set --request-file configs/srl2-bgp-srl3.json
{
  "source": "clab-dynamic-routing-bgp-srl2:57400",
  "timestamp": 1779672969484819948,
  "time": "2026-05-25T03:36:09.484819948+02:00",
  "results": [
    {
      "operation": "UPDATE",
      "path": "network-instance[name=default]/protocols/bgp/neighbor[peer-
address=10.1.6.2]"
    }
  ]
}

[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> gnmic -a clab-dynamic-routing-bgp-srl3:57400 -u admin -p NokiaSrl1! \
    --skip-verify -e json_ietf set --request-file configs/srl3-bgp-srl2.json
{
  "source": "clab-dynamic-routing-bgp-srl3:57400",
  "timestamp": 1779672978848160397,
  "time": "2026-05-25T03:36:18.848160397+02:00",
  "results": [
    {
      "operation": "UPDATE",
      "path": "network-instance[name=default]/protocols/bgp/neighbor[peer-
address=10.1.6.1]"
    }
  ]
}
```

4. Verify the new BGP session is established:

```
docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli -c "show network-instance default protocols bgp neighbor"
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli -c "show network-
instance default protocols bgp neighbor"
-----------------------------------------------------------------------------
----------------------------------------------------
BGP neighbor summary for network-instance "default"
Flags: S static, D dynamic, L discovered by LLDP, B BFD enabled, - disabled,
* slow
-----------------------------------------------------------------------------
----------------------------------------------------
-----------------------------------------------------------------------------
----------------------------------------------------
+--------------+---------------------+--------------+-----+--------+--------
----+------------+----------+---------------------+
|   Net-Inst   |        Peer         |    Group     | Fla | Peer-  |   State
|   Uptime   | AFI/SAFI |   [Rx/Active/Tx]    |
|              |                     |              | gs  |   AS   |
|            |          |                     |
+==============+=====================+==============+=====+========+========
====+============+==========+=====================+
| default      | 10.1.2.1            | ebgp-peers   | S   | 65001  |
establishe | 0d:0h:40m: | ipv4-    | [4/3/3]             |
|              |                     |              |     |        | d
| 23s        | unicast  |                     |
| default      | 10.1.6.2            | ebgp-peers   | S   | 65003  |
establishe | 0d:0h:0m:4 | ipv4-    | [0/0/0]             |
|              |                     |              |     |        | d
| s          | unicast  |                     |
+--------------+---------------------+--------------+-----+--------+--------
----+------------+----------+---------------------+
-----------------------------------------------------------------------------
----------------------------------------------------
Summary:
2 configured neighbors, 2 configured sessions are established, 0 disabled
peers
0 dynamic peers
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo',
'enter', 'environment', 'exit', 'file', 'help', 'history', 'info',
'interface', 'list', 'monitor', 'network-instance', 'passwd',
'ping', 'ping6', 'platform', 'pwc', 'qos', 'quit', 'references', 'routing-
policy', 'save', 'show', 'source', 'ssh', 'system', '
tcptraceroute', 'tech-support', 'tools', 'traceroute', 'traceroute6',
'tree', 'tunnel', 'tunnel-interface', 'watch', '{']
```

5. Traceroute AFTER -- host2 to host3 now goes direct (2 router hops):

```
docker exec clab-dynamic-routing-bgp-host2 traceroute -n -w 2 10.1.5.2
```

Expected: 10.1.4.1 (srl2) -> 10.1.6.2 (srl3) -> 10.1.5.2

6.
```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec clab-dynamic-routing-bgp-host2 traceroute -n -w 2 10.1.5.2
traceroute to 10.1.5.2 (10.1.5.2), 30 hops max, 60 byte packets
1  10.1.4.1  0.608 ms  0.611 ms  1.331 ms
2  10.1.6.2  1.129 ms  1.127 ms  1.175 ms
3  10.1.5.2  0.990 ms  0.997 ms  0.993 ms
```

7. Check received routes on srl2 to see the AS path difference:

```
docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli -c "show network-instance default protocols bgp routes ipv4 summary"
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli -c "show network-
instance default protocols bgp routes ipv4 summary"
-----------------------------------------------------------------------------
----------------------------------------------------
Show report for the BGP route table of network-instance "default"
-----------------------------------------------------------------------------
----------------------------------------------------
Status codes: u=used, *=valid, >=best, x=stale, b=backup
Origin codes: i=IGP, e=EGP, ?=incomplete
-----------------------------------------------------------------------------
----------------------------------------------------
+-----+-------------+-------------------+-----+-----+-------------------
--------------------+
| Sta |   Network    |       Next Hop        | MED | Loc |                Path
Val                 |
| tus |             |                   |     | Pre |
|
|     |             |                   |     |  f  |
|
+=====+=============+===================+=====+=====+===================
===================+
| u*> | 10.1.1.0/24  | 10.1.2.1             |     |     | [65001] i
|
| *   | 10.1.1.0/24  | 10.1.6.2             |     |     | [65003, 65001] i
|
| u*> | 10.1.2.0/24  | 0.0.0.0              |     |     |  i
|
| *   | 10.1.2.0/24  | 10.1.2.1             |     |     | [65001] i
|
| *   | 10.1.2.0/24  | 10.1.6.2             |     |     | [65003, 65001] i
|
| u*> | 10.1.3.0/24  | 10.1.2.1             |     |     | [65001] i
|
| *   | 10.1.3.0/24  | 10.1.6.2             |     |     | [65003] i
|
| u*> | 10.1.4.0/24  | 0.0.0.0              |     |     |  i
|
| *   | 10.1.5.0/24  | 10.1.2.1             |     |     | [65001, 65003] i
|
| u*> | 10.1.5.0/24  | 10.1.6.2             |     |     | [65003] i
|
| u*> | 10.1.6.0/24  | 0.0.0.0              |     |     |  i
|
| *   | 10.1.6.0/24  | 10.1.6.2             |     |     | [65003] i
|
```

```
+-----+-------------+-------------------+-----+-----+--------------------
--------------------+
-------------------------------------------------------------------------
----------------------------------------------------
12 received BGP routes: 6 used, 12 valid, 0 stale
6 available destinations: 6 with ECMP multipaths
-------------------------------------------------------------------------
----------------------------------------------------
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo',
'enter', 'environment', 'exit', 'file', 'help', 'history', 'info',
'interface', 'list', 'monitor', 'network-instance', 'passwd',
'ping', 'ping6', 'platform', 'pwc', 'qos', 'quit', 'references', 'routing-
policy', 'save', 'show', 'source', 'ssh', 'system', '
tcptraceroute', 'tech-support', 'tools', 'traceroute', 'traceroute6',
'tree', 'tunnel', 'tunnel-interface', 'watch', '{']
```

8. Inspect the best path detail for host3's subnet to see why BGP chose the direct path:

```
docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli -c "show network-
instance default protocols bgp routes ipv4 prefix 10.1.5.0/24 detail"
```

Look at the AS path for each received path. The best path algorithm checks Local Preference first (equal), then AS Path Length (shorter wins).

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli -c "show network-
instance default protocols bgp routes ipv4 prefix 10.
1.5.0/24 detail"
----------------------------------------------------------------------------
---------------------------------------------------
Show report for the BGP routes to network "10.1.5.0/24" network-instance
"default"
----------------------------------------------------------------------------
---------------------------------------------------
Network: 10.1.5.0/24
Received Paths: 2
 Path 1: <Valid,>
   Route source    : neighbor 10.1.2.1
   Route Preference: MED is -, No LocalPref
   BGP next-hop    : 10.1.2.1
   Path            :  i [65001, 65003]
   Communities     : None
   RR Attributes   : No Originator-ID, Cluster-List is [ - ]
   Aggregation     : Not an aggregate route
   Unknown Attr    : None
   Invalid Reason  : None
   Tie Break Reason: as-path-length
 Path 2: <Best,Valid,Used,>
   Route source    : neighbor 10.1.6.2
   Route Preference: MED is -, No LocalPref
   BGP next-hop    : 10.1.6.2
   Path            :  i [65003]
   Communities     : None
   RR Attributes   : No Originator-ID, Cluster-List is [ - ]
   Aggregation     : Not an aggregate route
   Unknown Attr    : None
   Invalid Reason  : None
   Tie Break Reason: none

Path 2 was advertised to:
[ 10.1.2.1 ]
Route Preference: MED is -, No LocalPref
Path            :  i [65002, 65003]
Communities     : None
RR Attributes   : No Originator-ID, Cluster-List is [ - ]
Aggregation     : Not an aggregate route
Unknown Attr    : None
----------------------------------------------------------------------------
---------------------------------------------------
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
```

```
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo',
'enter', 'environment', 'exit', 'file', 'help', 'history', 'info',
'interface', 'list', 'monitor', 'network-instance', 'passwd',
'ping', 'ping6', 'platform', 'pwc', 'qos', 'quit', 'references', 'routing-
policy', 'save', 'show', 'source', 'ssh', 'system', '
tcptraceroute', 'tech-support', 'tools', 'traceroute', 'traceroute6',
'tree', 'tunnel', 'tunnel-interface', 'watch', '{']
```

Note: In lesson 03, adding a cable accomplished nothing without adding static routes. With BGP, adding a cable and a peering session changes the forwarding path within seconds.

Deliverables

- Before/after traceroute output
- BGP route detail showing two paths with different AS path lengths
- Explanation of which best path algorithm step decided the winner
  - *BGP's best path algorithm selected the direct path because it has a shorter AS path. In this lab, AS Path Length is the deciding factor because all other attributes (Local Preference, Origin, MED) are equal. In production networks, operators use Local Preference and MED to influence path selection beyond what AS path length provides.* (from solution)

✅ done

Exercise 3: Break/Fix -- Missing Export Policy

Objective: Understand that a BGP session being Established does not mean routes are flowing.

Setup (break it)

```
docker exec -it clab-dynamic-routing-bgp-srl3 sr_cli
```

Inside SR Linux:

```
enter candidate
delete / network-instance default protocols bgp group ebgp-peers export-
policy
commit now
exit
```

Symptom

```
# host1 and host2 CANNOT reach host3
docker exec clab-dynamic-routing-bgp-host1 ping -c 3 -W 5 10.1.5.2
docker exec clab-dynamic-routing-bgp-host2 ping -c 3 -W 5 10.1.5.2

# But host3 CAN still reach host1 and host2 (it still receives routes)
docker exec clab-dynamic-routing-bgp-host3 ping -c 3 10.1.1.2
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec clab-dynamic-routing-bgp-host1 ping -c 3 -W 5 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
From 10.1.1.1 icmp_seq=1 Destination Net Unreachable
From 10.1.1.1 icmp_seq=2 Destination Net Unreachable
From 10.1.1.1 icmp_seq=3 Destination Net Unreachable

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2040ms


[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec clab-dynamic-routing-bgp-host2 ping -c 3 -W 5 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
From 10.1.4.1 icmp_seq=1 Destination Net Unreachable
From 10.1.4.1 icmp_seq=2 Destination Net Unreachable
From 10.1.4.1 icmp_seq=3 Destination Net Unreachable

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2033ms


[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec clab-dynamic-routing-bgp-host3 ping -c 3 10.1.1.2
PING 10.1.1.2 (10.1.1.2) 56(84) bytes of data.

--- 10.1.1.2 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2040ms
```

Your Task

1. Check BGP neighbor detail on srl3:

   ```
   docker exec -it clab-dynamic-routing-bgp-srl3 sr_cli -c "show network-
   instance default protocols bgp neighbor 10.1.3.1 detail"
   ```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl3 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl3# show network-instance default protocols bgp neighbor 10.1.3.1 detail
-------------------------------------------------------------------------------
----------------------------------------------------
Peer : 10.1.3.1, remote AS: 65001, local AS: 65003, peer-type : ebgp
Type : static
Description : None
Group : ebgp-peers
Export policies : ['export-connected', 'export-bgp']
Import policies: ['import-all']
Under maintenance: False
Maintenance group:
-------------------------------------------------------------------------------
----------------------------------------------------
Admin-state is enable, session-state is established, up for 0d:0h:50m:12s
TCP connection is 10.1.3.2 [179] -> 10.1.3.1 [37401]
TCP-MD5 authentication is disabled
0 messages in input queue, 0 messages in output queue
-------------------------------------------------------------------------------
----------------------------------------------------
Last-state was active, last-event was recvOpen, 1 peer-flaps
Last received Notification was Error: None SubError: None
Failure detection: BFD is False, fast-failover is True
-------------------------------------------------------------------------------
----------------------------------------------------
Graceful Restart
  Admin State : disable
  Restarts by the peer : None
  Last restart : N/A
  Peer requested restart-time : None
  Stale routes time : None
-------------------------------------------------------------------------------
----------------------------------------------------
               Timer                        Configured
Operational                        Next
===============================================================================
===============================================
connect-retry                                               120                     120
-
keepalive-interval                                          30                      30
-
```

```
hold-time                                                90                     90
-
minimum-advertisement-interval                            5                      5
-
prefix-limit-restart-timer                                0                      0
-
-------------------------------------------------------------------------------
----------------------------------------------------
Cap Sent:  ROUTE_REFRESH 4-OCTET_ASN MP_BGP GRACEFUL_RESTART
Cap Recv:  ROUTE_REFRESH 4-OCTET_ASN MP_BGP GRACEFUL_RESTART
-------------------------------------------------------------------------------
----------------------------------------------------
            Messages                                        Sent
Received                   Last
===============================================================================
================================================
Non Updates                                                   103
103
Updates                                                         7
6                    2026-05-

25T01:40:

20.300Z
Malformed updates                                               0
0
Route Refreshes                                                 0
0
-------------------------------------------------------------------------------
----------------------------------------------------
Ipv4-unicast AFI/SAFI
   End of RIB                      : sent, received
   Received routes                 : 5
   Rejected routes                 : None
   Active routes                   : 2
   Advertised routes               : 4
   Prefix-limit-received           : 4294967295 routes, warning at 90,
prevent-teardown False
   Prefix-limit-accepted           : None
   Default originate               : disabled
   Advertise with IPv6 next-hops   : False
   Peer requested GR helper        : None
   Peer preserved forwarding state: None
-------------------------------------------------------------------------------
----------------------------------------------------
--{ + running }--[  ]--
```

> ⚠ **Attention**
> I don't know why, but I get other output in my environment with the same command on `srl3`

2. Look at received vs sent route counts. srl3 receives routes (it can reach out) but sends 0 (nobody can reach it).
3. Fix by re-adding both export policies:

   ```
   docker exec -it clab-dynamic-routing-bgp-srl3 sr_cli
   ```

   Inside SR Linux:

   ```
   enter candidate
   set / network-instance default protocols bgp group ebgp-peers export-policy [export-connected export-bgp]
   commit now
   exit
   ```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl3 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl3# enter candidate
    set / network-instance default protocols bgp group ebgp-peers export-policy [export-connected export-bgp]
    commit now
    exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
A:srl3#
EOF encountered
```

4. Verify:

   ```
   docker exec clab-dynamic-routing-bgp-host1 ping -c 3 10.1.5.2
   ```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec clab-dynamic-routing-bgp-host1 ping -c 3 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
64 bytes from 10.1.5.2: icmp_seq=1 ttl=62 time=0.448 ms
64 bytes from 10.1.5.2: icmp_seq=2 ttl=62 time=0.359 ms
64 bytes from 10.1.5.2: icmp_seq=3 ttl=62 time=0.415 ms

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2052ms
rtt min/avg/max/mdev = 0.359/0.407/0.448/0.036 ms
```

Deliverables

- BGP neighbor detail showing sent-routes=0
- Explanation of SR Linux default-deny export behavior

✅ done

Exercise 4: Break/Fix -- Link Failure with Automatic Reroute

Objective: Observe dynamic routing self-healing -- the same break from lesson 03 exercise 6 that was permanent now fixes itself.

Setup (break it)

```
docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / interface ethernet-1/3 admin-state disable
commit now
exit
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl1# enter candidate
set / interface ethernet-1/3 admin-state disable
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
A:srl1#
```

Symptom

```
# Brief connectivity loss to host3, then recovery
# Run continuous ping to watch:
docker exec clab-dynamic-routing-bgp-host1 ping -c 30 -i 1 10.1.5.2
```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec clab-dynamic-routing-bgp-host1 ping -c 30 -i 1 10.1.5.2
#PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
64 bytes from 10.1.5.2: icmp_seq=1 ttl=61 time=0.505 ms
64 bytes from 10.1.5.2: icmp_seq=2 ttl=61 time=0.582 ms
64 bytes from 10.1.5.2: icmp_seq=3 ttl=61 time=0.629 ms
64 bytes from 10.1.5.2: icmp_seq=4 ttl=61 time=0.494 ms
64 bytes from 10.1.5.2: icmp_seq=5 ttl=61 time=0.613 ms
^C
```

Your Task

1. While the continuous ping is running, watch for packet loss then recovery.
2. Check srl1's BGP neighbors -- the srl3 session should drop:

   ```
   docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli -c "show network-
   instance default protocols bgp neighbor"
   ```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli -c "show network-
instance default protocols bgp neighbor"
-----------------------------------------------------------------------------
----------------------------------------------------
BGP neighbor summary for network-instance "default"
Flags: S static, D dynamic, L discovered by LLDP, B BFD enabled, - disabled,
* slow
-----------------------------------------------------------------------------
----------------------------------------------------
-----------------------------------------------------------------------------
----------------------------------------------------
+-------------+---------------------+-------------+-----+--------+--------
----+-----------+----------+--------------------+
|   Net-Inst   |        Peer         |    Group     | Fla | Peer-  |   State
|   Uptime   | AFI/SAFI |   [Rx/Active/Tx]    |
|              |                     |              | gs  |   AS   |
|            |          |                      |
+=============+=====================+=============+=====+========+========
====+===========+==========+====================+
| default      | 10.1.2.1            | ebgp-peers   | S   | 65001  | active
| -          |          |                      |
| default      | 10.1.2.2            | ebgp-peers   | S   | 65002  |
establishe | 0d:1h:1m:6 | ipv4-    | [4/3/2]              |
|              |                     |              |     |        | d
| s          | unicast  |                      |
| default      | 10.1.3.2            | ebgp-peers   | S   | 65003  | active
| -          |          |                      |
+-------------+---------------------+-------------+-----+--------+--------
----+-----------+----------+--------------------+
-----------------------------------------------------------------------------
----------------------------------------------------
Summary:
3 configured neighbors, 1 configured sessions are established, 0 disabled
peers
0 dynamic peers
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo',
'enter', 'environment', 'exit', 'file', 'help', 'history', 'info',
'interface', 'list', 'monitor', 'network-instance', 'passwd',
'ping', 'ping6', 'platform', 'pwc', 'qos', 'quit', 'references', 'routing-
policy', 'save', 'show', 'source', 'ssh', 'system', '
tcptraceroute', 'tech-support', 'tools', 'traceroute', 'traceroute6',
'tree', 'tunnel', 'tunnel-interface', 'watch', '{']
```

3. Traceroute to see the new path (traffic now goes srl1 -> srl2 -> srl3 instead of direct):

```
docker exec clab-dynamic-routing-bgp-host1 traceroute -n -w 2 10.1.5.2
```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec clab-dynamic-routing-bgp-host1 traceroute -n -w 2 10.1.5.2
traceroute to 10.1.5.2 (10.1.5.2), 30 hops max, 60 byte packets
1  10.1.1.1  1.854 ms  2.179 ms  2.177 ms
2  10.1.2.2  1.804 ms  1.803 ms  1.801 ms
3  10.1.6.2  3.066 ms  3.065 ms  3.065 ms
4  10.1.5.2  2.468 ms  2.951 ms  3.002 ms
```

3. Re-enable the link:

```
docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / interface ethernet-1/3 admin-state enable
commit now
exit
```

4. Watch the BGP session come back up and traffic shift back to the direct path.

Callback: In lesson 03, disabling this same link permanently broke connectivity to host3. Now with BGP, the network found an alternate path automatically.

Deliverables

- Ping output showing loss then recovery
- Before/after traceroute
- Explanation of BGP convergence

✅ done

Exercise 5: Break/Fix -- Wrong ASN

Objective: Understand what happens when BGP peers disagree on AS numbers.

Setup (break it)

```
docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli
```

Inside SR Linux:

```
enter candidate
set / network-instance default protocols bgp neighbor 10.1.2.1 peer-as 65099
commit now
exit
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl2# enter candidate
set / network-instance default protocols bgp neighbor 10.1.2.1 peer-as 65099
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
A:srl2# docker exec clab-dynamic-routing-bgp-host2 ping -c 3 -W 5 10.1.1.2
--{ + running }--[  ]--
A:srl2#
EOF encountered
```

Symptom

```
# Pings through srl2 fail (srl2 loses routes from srl1)
docker exec clab-dynamic-routing-bgp-host2 ping -c 3 -W 5 10.1.1.2
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec clab-dynamic-routing-bgp-host2 ping -c 3 -W 5 10.1.1.2
PING 10.1.1.2 (10.1.1.2) 56(84) bytes of data.
64 bytes from 10.1.1.2: icmp_seq=1 ttl=61 time=1.17 ms
64 bytes from 10.1.1.2: icmp_seq=2 ttl=61 time=1.75 ms
64 bytes from 10.1.1.2: icmp_seq=3 ttl=61 time=0.977 ms

--- 10.1.1.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.977/1.298/1.749/0.328 ms
```

Your Task

1. Check BGP neighbors on srl2 -- the srl1 session should be stuck in `active` or `connect`:

   ```
   docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli -c "show network-
   instance default protocols bgp neighbor"
   ```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli -c "show network-
instance default protocols bgp neighbor"
-----------------------------------------------------------------------------
----------------------------------------------------
BGP neighbor summary for network-instance "default"
Flags: S static, D dynamic, L discovered by LLDP, B BFD enabled, - disabled,
* slow
-----------------------------------------------------------------------------
----------------------------------------------------
-----------------------------------------------------------------------------
----------------------------------------------------
+--------------+---------------------+--------------+-----+--------+--------
----+------------+----------+---------------------+
|   Net-Inst   |        Peer         |    Group     | Fla | Peer-  |   State
|   Uptime   | AFI/SAFI |   [Rx/Active/Tx]    |
|              |                     |              | gs  |   AS   |
|            |          |                     |
+==============+=====================+==============+=====+========+========
====+============+==========+=====================+
| default      | 10.1.2.1            | ebgp-peers   | S   | 65099  | active
| -          |          |                     |
| default      | 10.1.6.2            | ebgp-peers   | S   | 65003  |
establishe | 0d:0h:23m: | ipv4-    | [5/3/3]             |
|              |                     |              |     |        | d
| 40s        | unicast  |                     |
+--------------+---------------------+--------------+-----+--------+--------
----+------------+----------+---------------------+
-----------------------------------------------------------------------------
----------------------------------------------------
Summary:
2 configured neighbors, 1 configured sessions are established, 0 disabled
peers
0 dynamic peers
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo',
'enter', 'environment', 'exit', 'file', 'help', 'history', 'info',
'interface', 'list', 'monitor', 'network-instance', 'passwd',
'ping', 'ping6', 'platform', 'pwc', 'qos', 'quit', 'references', 'routing-
policy', 'save', 'show', 'source', 'ssh', 'system', '
tcptraceroute', 'tech-support', 'tools', 'traceroute', 'traceroute6',
'tree', 'tunnel', 'tunnel-interface', 'watch', '{']
```

2. Check srl1's view -- it expects AS 65002, but srl2 now announces itself as expecting AS 65099.

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli -c "show network-
instance default protocols bgp neighbor"
-----------------------------------------------------------------------------
----------------------------------------------------
BGP neighbor summary for network-instance "default"
Flags: S static, D dynamic, L discovered by LLDP, B BFD enabled, - disabled,
* slow
-----------------------------------------------------------------------------
----------------------------------------------------
-----------------------------------------------------------------------------
----------------------------------------------------
+--------------+---------------------+--------------+-----+--------+--------
----+------------+----------+---------------------+
|   Net-Inst   |        Peer         |    Group     | Fla | Peer-  |   State
|   Uptime   | AFI/SAFI |   [Rx/Active/Tx]    |
|              |                     |              | gs  |   AS   |
|            |          |                     |
+==============+=====================+==============+=====+========+========
====+============+==========+=====================+
| default      | 10.1.2.1            | ebgp-peers   | S   | 65001  | active
| -          |          |                     |
| default      | 10.1.2.2            | ebgp-peers   | S   | 65002  | active
| -          |          |                     |
| default      | 10.1.3.2            | ebgp-peers   | S   | 65003  |
establishe | 0d:0h:1m:5 | ipv4-    | [4/3/3]             |
|              |                     |              |     |        | d
| 8s         | unicast  |                     |
+--------------+---------------------+--------------+-----+--------+--------
----+------------+----------+---------------------+
-----------------------------------------------------------------------------
----------------------------------------------------
Summary:
3 configured neighbors, 1 configured sessions are established, 0 disabled
peers
0 dynamic peers
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo',
'enter', 'environment', 'exit', 'file', 'help', 'history', 'info',
'interface', 'list', 'monitor', 'network-instance', 'passwd',
'ping', 'ping6', 'platform', 'pwc', 'qos', 'quit', 'references', 'routing-
policy', 'save', 'show', 'source', 'ssh', 'system', '
tcptraceroute', 'tech-support', 'tools', 'traceroute', 'traceroute6',
'tree', 'tunnel', 'tunnel-interface', 'watch', '{']
```

3. Fix by restoring the correct peer-as:

```
docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli
```

Inside SR Linux:

```
enter candidate
set / network-instance default protocols bgp neighbor 10.1.2.1 peer-as
65001
commit now
exit
```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl2# enter candidate
  set / network-instance default protocols bgp neighbor 10.1.2.1 peer-as
65001
  commit now
  exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
```

4. Verify:

```
docker exec clab-dynamic-routing-bgp-host2 ping -c 3 10.1.1.2
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──>  docker exec clab-dynamic-routing-bgp-host2 ping -c 3 10.1.1.2
PING 10.1.1.2 (10.1.1.2) 56(84) bytes of data.
64 bytes from 10.1.1.2: icmp_seq=1 ttl=62 time=0.889 ms
64 bytes from 10.1.1.2: icmp_seq=2 ttl=62 time=0.365 ms
64 bytes from 10.1.1.2: icmp_seq=3 ttl=62 time=0.381 ms

--- 10.1.1.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2027ms
rtt min/avg/max/mdev = 0.365/0.545/0.889/0.243 ms
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl2 sr_cli -c \
   "show network-instance default protocols bgp neighbor"
----------------------------------------------------------------------------
----------------------------------------------------
BGP neighbor summary for network-instance "default"
Flags: S static, D dynamic, L discovered by LLDP, B BFD enabled, - disabled,
* slow
----------------------------------------------------------------------------
----------------------------------------------------
----------------------------------------------------------------------------
----------------------------------------------------
+-------------+-------------------+-------------+-----+--------+--------
----+-----------+---------+-------------------+
|   Net-Inst   |        Peer        |    Group     | Fla | Peer-  |   State
|   Uptime   | AFI/SAFI |   [Rx/Active/Tx]    |
|             |                   |             | gs  |   AS   |
|           |         |                   |
+=============+===================+=============+=====+========+========
====+===========+=========+===================+
| default      | 10.1.2.1           | ebgp-peers   | S   | 65001  |
establishe | 0d:0h:14m: | ipv4-    | [4/2/4]             |
|             |                   |             |     |        | d
| 7s         | unicast  |                   |
| default      | 10.1.6.2           | ebgp-peers   | S   | 65003  |
establishe | 0d:0h:39m: | ipv4-    | [5/1/5]             |
|             |                   |             |     |        | d
| 9s         | unicast  |                   |
+-------------+-------------------+-------------+-----+--------+--------
----+-----------+---------+-------------------+
----------------------------------------------------------------------------
----------------------------------------------------
Summary:
2 configured neighbors, 2 configured sessions are established, 0 disabled
peers
0 dynamic peers
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo',
'enter', 'environment', 'exit', 'file', 'help', 'history', 'info',
'interface', 'list', 'monitor', 'network-instance', 'passwd',
'ping', 'ping6', 'platform', 'pwc', 'qos', 'quit', 'references', 'routing-
policy', 'save', 'show', 'source', 'ssh', 'system', '
tcptraceroute', 'tech-support', 'tools', 'traceroute', 'traceroute6',
'tree', 'tunnel', 'tunnel-interface', 'watch', '{']
```

Deliverables

- BGP neighbor output showing non-established state
- Explanation of AS mismatch
- **From Drew:** *Both sides must agree on AS numbers. When debugging BGP, check session state first. If a session is stuck in `active` or `connect`, the most likely causes are: wrong peer-as, wrong peer-address, or a firewall blocking TCP port 179. The `active` state specifically suggests the TCP connection succeeds but the OPEN message is rejected -- which points to an AS number mismatch.*

✅ done

Exercise 6: Break/Fix -- Stale Static Route Masks BGP

Objective: Understand administrative distance -- why a manually added static route overrides a correct BGP-learned route.

Setup (break it)

```
docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / network-instance default next-hop-groups group nhg-wrong admin-state
enable
set / network-instance default next-hop-groups group nhg-wrong nexthop 1 ip-
address 10.1.3.2
set / network-instance default static-routes route 10.1.4.0/24 admin-state
enable
set / network-instance default static-routes route 10.1.4.0/24 next-hop-
group nhg-wrong
commit now
exit
```

This adds a static route on srl1 for host2's subnet (10.1.4.0/24) pointing at srl3 (10.1.3.2) -- the wrong direction. BGP correctly points this prefix at srl2 (10.1.2.2).

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl1# enter candidate
set / network-instance default next-hop-groups group nhg-wrong admin-state
enable
set / network-instance default next-hop-groups group nhg-wrong nexthop 1 ip-
address 10.1.3.2
set / network-instance default static-routes route 10.1.4.0/24 admin-state
enable
set / network-instance default static-routes route 10.1.4.0/24 next-hop-
group nhg-wrong
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
```

Symptom

```
# host1 CANNOT reach host2
docker exec clab-dynamic-routing-bgp-host1 ping -c 3 -W 5 10.1.4.2

# But host1 CAN still reach host3
docker exec clab-dynamic-routing-bgp-host1 ping -c 3 10.1.5.2
```

```
$ docker exec clab-dynamic-routing-bgp-host1 ping -c 3 -W 5 10.1.4.2
PING 10.1.4.2 (10.1.4.2): 56 data bytes

--- 10.1.4.2 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss

$ docker exec clab-dynamic-routing-bgp-host1 ping -c 3 10.1.5.2
PING 10.1.5.2 (10.1.5.2): 56 data bytes
64 bytes from 10.1.5.2: seq=0 ttl=62 time=2.345 ms
64 bytes from 10.1.5.2: seq=1 ttl=62 time=1.123 ms
64 bytes from 10.1.5.2: seq=2 ttl=62 time=0.987 ms
--- 10.1.5.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
```

Your Task

1. Check the routing table on srl1:

```
docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli -c "show network-
instance default route-table ipv4-unicast summary"
```

```
--{ + running }--[  ]--
A:srl1# show network-instance default route-table ipv4-unicast summary |
grep -i "prefix\|static\|local\|bgp"
| Prefix  |   ID     |  Route   |  Route   | Active   | Origin   | Metric   |
Pref    |  Next-   |  Next-   | Backup   | Backup   |
| 10.1.1. | 2        | local    | net_ins | True     | default | 0        | 0
| 10.1.1. | etherne |          |          |
| 10.1.2. | 3        | local    | net_ins | True     | default | 0        | 0
| 10.1.2. | etherne |          |          |
| 10.1.3. | 4        | local    | net_ins | True     | default | 0        | 0
| 10.1.3. | etherne |          |          |
| 10.1.4. | 0        | static   | static_ | True     | default | 1        | 5
| 10.1.3. | etherne |          |          |
|         |          |          |          |          |          |          |
| /local) |          |          |          |
| 10.1.5. | 0        | bgp      | bgp_mgr | True     | default | 0        | 170
| 10.1.3. | etherne |          |          |
|         |          |          |          |          |          |          |
| /local) |          |          |          |
| 10.1.6. | 0        | bgp      | bgp_mgr | True     | default | 0        | 170
| 10.1.2. | etherne |          |          |
|         |          |          |          |          |          |          |
| /local) |          |          |          |
IPv4 prefixes with active routes     : 12
IPv4 prefixes with active ECMP routes: 0

--{ + running }--[  ]--
```

2. Look at the route type for 10.1.4.0/24. It should show `static` instead of `bgp` -- even though BGP is still running and the session is healthy.
3. Check BGP neighbors on srl1 -- sessions are all `established`:

```
docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli -c "show network-
instance default protocols bgp neighbor"
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli -c "show network-
instance default protocols bgp neighbor"
-----------------------------------------------------------------------------
----------------------------------------------------
BGP neighbor summary for network-instance "default"
Flags: S static, D dynamic, L discovered by LLDP, B BFD enabled, - disabled,
* slow
-----------------------------------------------------------------------------
----------------------------------------------------
-----------------------------------------------------------------------------
----------------------------------------------------
+-------------+---------------------+-------------+-----+--------+--------
----+-----------+---------+--------------------+
|   Net-Inst   |        Peer         |    Group     | Fla | Peer-  |   State
|   Uptime   | AFI/SAFI |   [Rx/Active/Tx]    |
|              |                     |              | gs  |   AS   |
|            |          |                     |
+=============+=====================+=============+=====+========+========
====+===========+=========+====================+
| default      | 10.1.2.1            | ebgp-peers   | S   | 65001  | active
| -          |          |                     |
| default      | 10.1.2.2            | ebgp-peers   | S   | 65002  |
establishe | 0d:0h:19m: | ipv4-    | [4/1/4]            |
|              |                     |              |     |        | d
| 1s         | unicast  |                     |
| default      | 10.1.3.2            | ebgp-peers   | S   | 65003  |
establishe | 0d:0h:21m: | ipv4-    | [4/1/4]            |
|              |                     |              |     |        | d
| 34s        | unicast  |                     |
+-------------+---------------------+-------------+-----+--------+--------
----+-----------+---------+--------------------+
-----------------------------------------------------------------------------
----------------------------------------------------
Summary:
3 configured neighbors, 2 configured sessions are established, 0 disabled
peers
0 dynamic peers
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo',
'enter', 'environment', 'exit', 'file', 'help', 'history', 'info',
'interface', 'list', 'monitor', 'network-instance', 'passwd',
'ping', 'ping6', 'platform', 'pwc', 'qos', 'quit', 'references', 'routing-
policy', 'save', 'show', 'source', 'ssh', 'system', '
tcptraceroute', 'tech-support', 'tools', 'traceroute', 'traceroute6',
'tree', 'tunnel', 'tunnel-interface', 'watch', '{']
```

4. Fix by deleting the stale static route:

```
docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
delete / network-instance default static-routes route 10.1.4.0/24
delete / network-instance default next-hop-groups group nhg-wrong
commit now
exit
```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec -it clab-dynamic-routing-bgp-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl1#    enter candidate
  delete / network-instance default static-routes route 10.1.4.0/24
  delete / network-instance default next-hop-groups group nhg-wrong
  commit now
  exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
A:srl1#
EOF encountered
```

5. Verify:

```
docker exec clab-dynamic-routing-bgp-host1 ping -c 3 10.1.4.2
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp/gnmic]
└──> docker exec clab-dynamic-routing-bgp-host1 ping -c 3 10.1.4.2
PING 10.1.4.2 (10.1.4.2) 56(84) bytes of data.
64 bytes from 10.1.4.2: icmp_seq=1 ttl=62 time=0.536 ms
64 bytes from 10.1.4.2: icmp_seq=2 ttl=62 time=0.319 ms
64 bytes from 10.1.4.2: icmp_seq=3 ttl=62 time=0.316 ms

--- 10.1.4.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2056ms
rtt min/avg/max/mdev = 0.316/0.390/0.536/0.103 ms
```

Deliverables

- Routing table showing `static` type for 10.1.4.0/24 with the wrong next-hop
- Explanation of why static routes (admin distance 5) beat BGP routes (admin distance 170)

by Drew

The critical clue: 10.1.4.0/24 shows `static` with next-hop 10.1.3.2 (srl3). But host2 is behind srl2, not srl3. The correct next-hop should be 10.1.2.2. BGP knows this -- but the static route wins.

Why the static route wins: Every routing source has an administrative distance -- a trust level that determines priority when multiple sources provide a route for the same prefix. On SR Linux:

Route Source	Administrative Distance
Local/connected	0
Static	5
BGP	170

Lower distance wins. The static route (distance 5) beats the BGP route (distance 170) for 10.1.4.0/24, even though the BGP route has the correct next-hop. The BGP route is still learned and stored, but it is hidden -- the routing table installs the static route because it is more "trusted."

This is one of the most common production gotchas when migrating from static to dynamic routing. Old static routes left behind will silently override BGP-learned routes. Everything looks healthy -- BGP sessions are established, routes are being exchanged -- but traffic goes to the wrong place because a stale static route has priority.

BGP neighbors on srl1 (all healthy -- this is the trap):

```
A:srl1# show network-instance default protocols bgp neighbor
+---------------+--------+--------+-----------+--------+--------+--------+
| Peer          | Group  | AS     | Admin     | Session| Rcv    | Active |
|               |        |        | State     | State  | Routes | Routes |
+===============+========+========+===========+========+========+========+
| 10.1.2.2      |ebgp-pe | 65002  | enable    |establis| 2      | 1      |
|               |  ers   |        |           | hed    |        |        |
+---------------+--------+--------+-----------+--------+--------+--------+
| 10.1.3.2      |ebgp-pe | 65003  | enable    |establis| 2      | 2      |
|               |  ers   |        |           | hed    |        |        |
+---------------+--------+--------+-----------+--------+--------+--------+
```

Notice that srl2's session shows `Rcv: 2` but `Active: 1`. srl1 received 2 routes from srl2 (10.1.2.0/24 and 10.1.4.0/24), but only 1 is active -- 10.1.4.0/24 was received but not installed because the static route has a lower administrative distance. This `Rcv > Active` mismatch is the diagnostic clue that another routing source is overriding BGP.

Cleanup

After completing all exercises:

```
# Destroy the lab
containerlab destroy -t topology/lab.clab.yml --cleanup

# Verify
docker ps | grep clab
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp]
└──> containerlab destroy -t topology/lab.clab.yml --cleanup
04:23:24 INFO Parsing & checking topology file=lab.clab.yml
04:23:24 INFO Parsing & checking topology file=lab.clab.yml
04:23:25 INFO Destroying lab name=dynamic-routing-bgp
04:23:25 INFO Removed container name=clab-dynamic-routing-bgp-host1
04:23:25 INFO Removed container name=clab-dynamic-routing-bgp-host2
04:23:25 INFO Removed container name=clab-dynamic-routing-bgp-host3
04:23:25 INFO Removed container name=clab-dynamic-routing-bgp-srl3
04:23:25 INFO Removed container name=clab-dynamic-routing-bgp-srl1
04:23:25 INFO Removed container name=clab-dynamic-routing-bgp-srl2
04:23:25 INFO Removing host entries path=/etc/hosts
04:23:25 INFO Removing SSH config path=/etc/ssh/ssh_config.d/clab-dynamic-
routing-bgp.conf

[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/04-dynamic-routing-bgp]
└──> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

Validation

Run the automated tests:

```
pytest tests/ -v
```

Test fails because I destroyed the infrastructure right before I executed the test. This did not happen in the lessons/excersises before :-/

Completion Checklist

- ✅ Exercise 1: Deployed lab, verified broken state, configured eBGP, compared routing tables
- ✅ Exercise 2: Enabled direct link, observed AS path selection via traceroute
- ✅ Exercise 3: Diagnosed and fixed missing export policy (sent-routes=0)
- ✅ Exercise 4: Observed link failure with automatic reroute (BGP convergence)
- ✅ Exercise 5: Diagnosed and fixed wrong ASN (peer-as mismatch)
- ✅ Exercise 6: Diagnosed stale static route masking BGP (administrative distance)

Next Steps

Commit your exercises to your fork and proceed to Lesson 5 (coming soon).